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FA23-BAI-042

Field Work in AI/ML

V1 – Getting Started with AI/ML

In the first week of my AI/ML journey, I focused on setting up my working environment and learning the basics. I installed Python using Anaconda and worked with Jupyter Notebook. I explored libraries like NumPy and Pandas, which helped me understand how to handle and organize data.

I also practiced data cleaning things like removing missing values, converting text into numbers, and scaling the data so that it's easier for a machine to understand. At this stage, I didn't use any advanced machine learning libraries. Instead, I tried to build a basic understanding of how data flows and how machines can make decisions using simple logic.

This week gave me a solid starting point and helped me realize how important clean data is before doing anything with AI.

End of V1.

V2 – Exploring Data: Titanic Dataset EDA

In the second week, I worked on a very famous dataset the Titanic dataset. My goal was to explore the data and understand the patterns behind who survived and who didn't.

I used Pandas, Matplotlib, and Seaborn to visualize the data. I made graphs that showed how survival was related to gender, age, ticket class, and other features. One thing that stood out was that women and first-class passengers had higher survival rates the visuals really helped me understand this better.

This process of **EDA (Exploratory Data Analysis)** taught me how important it is to explore and understand the data before jumping into any modeling. I learned how to ask the right questions, spot patterns, and prepare data for future machine learning tasks.

It was a fun and interesting way to let the data tell its story and I really started to enjoy the process.

End of V2.

V3 – Supervised Learning: Regression & Classification

In my third week of the AI/ML internship, I stepped into the world of **supervised learning** where the machine is trained using labeled data to make predictions.

I started with **Linear Regression** using the Boston Housing dataset. The goal was to predict house prices based on features like the number of rooms, tax rate, and location. I learned how to split the data into training and testing sets, train the model using `fit()`, and then evaluate it using metrics like **Mean Squared Error (MSE)** and **R^2 Score**. It was exciting to see how close my model could predict real housing prices!

Next, I used **Logistic Regression** on the Titanic dataset to predict whether a passenger survived or not. This time, instead of predicting a number, the model had to predict a **Yes/No** outcome. I applied data preprocessing techniques, handled missing values, converted text to numbers (like ‘male’ to 0 and ‘female’ to 1), and then trained the model. I also learned how to evaluate classification models using **accuracy** and **confusion matrices**, which helped me understand how well my model was performing.

This week taught me the difference between regression and classification, how to train models with labeled data, and how to interpret results. I finally saw my models making real predictions and it made me even more curious to explore more complex algorithms in the coming weeks.

End of V3.

V4 – Unsupervised Learning: Clustering with K-Means and PCA

In Week 4, I explored the other side of machine learning **unsupervised learning**, where the model learns patterns from data without any labels.

I worked with the **Iris dataset**, which contains measurements of three types of flowers. But this time, I didn’t tell the model what the flower types were I let it figure it out on its own using **K-Means Clustering**.

K-Means is a way to group similar data points together into clusters. I set the number of clusters to 3 (since I knew there were three flower types) and trained the model. It was interesting to see that even without being told the correct labels, the algorithm was able to group the flowers quite accurately.

To visualize the results, I used **PCA (Principal Component Analysis)**, which reduces the number of features while keeping the most important information. This allowed me to create a 2D plot of the clusters and it made the results much easier to understand.

This week taught me that even without labels, machine learning can find structure in data. Unsupervised learning feels a bit like data exploration and pattern discovery, which makes it powerful for real-world tasks where labeled data isn't always available.

End of V4.

V5 – Time-Series Anomaly Detection (Stock Market Project)

This week, I explored how AI can be used to detect **unusual behavior in stock prices** which is important in finance for spotting market manipulation or unexpected trends.

I downloaded historical stock data using Yahoo Finance and calculated technical indicators like **SMA (Simple Moving Average)**, **RSI (Relative Strength Index)**, and **Bollinger Bands**. These indicators helped me understand how traders analyze price movements.

Then, I applied **Isolation Forest**, an unsupervised algorithm that identifies anomalies in time-series data. I also experimented with **Prophet**, a forecasting tool from Meta, to predict future trends and detect points where the real data started behaving differently than expected.

This task taught me how machine learning can spot things humans might miss and how important time and trend-based data is in real-world applications like finance.

End of V5.

V6 – Multi-Label Emotion Recognition with BERT

In Week 6, I worked on a very interesting problem identifying **multiple emotions in a single sentence**.

I used the **GoEmotions dataset** by Google, which includes text examples labeled with emotions like joy, anger, sadness, and more. Since each sentence could have more than one emotion, this became a **multi-label classification** task.

I fine-tuned a **BERT transformer model**, which is one of the most powerful tools in Natural Language Processing (NLP). I trained it to read sentences and detect all the emotions present. For example, the sentence "I'm happy but nervous" could be classified as both joy and fear.

This week helped me understand how **deep learning models like BERT** work with text, and how AI can actually interpret emotional tone in a more human way.

End of V6.

V7 – Satellite Image Analysis for Deforestation

This week was all about **computer vision and sustainability**. I used satellite images to detect **deforestation** over time in the Amazon rainforest.

I preprocessed multi-band satellite images and trained a **Convolutional Neural Network (CNN)** to classify whether a piece of land was forest or deforested. I also compared images from different time periods to identify where forests were being cut down.

I visualized the results by marking the deforested areas on the satellite map. This made me realize how **AI can help in environmental monitoring**, and how visual data (like images from space) can be used to track serious changes on Earth.

End of V7.

V8 – Credit Risk Prediction

In Week 8, I moved to the world of **finance and credit analysis**. The goal was to predict whether a person was at risk of defaulting on a loan.

I used the **Give Me Some Credit dataset**, which had information like income, age, number of dependents, and previous payment history. I cleaned the data and handled class imbalance using **SMOTE** (a technique to balance the dataset).

Then, I trained models like **Random Forest** and **XGBoost** to predict credit risk. I evaluated the models using metrics like **ROC-AUC** and **confusion matrix**.

This project helped me understand how banks and financial institutions use machine learning to **make smarter lending decisions** and reduce risk.

End of V8.

V9 – Model Evaluation and Fine-Tuning

This week, I focused on **improving model performance**. I explored concepts like:

- **Hyperparameter tuning** (changing settings like learning rate or tree depth)

- **Cross-validation** to test models more reliably
- **Feature selection** to remove unnecessary or less important inputs

I learned that building a model is only half the job making it better and more reliable is just as important. I compared models using metrics like **F1-score**, **precision**, and **recall**, and plotted things like **learning curves** to see how models were improving.

This helped me understand that **good results come from both smart design and careful testing**.

End of V9.

V10 – Internship Wrap-Up & Final Presentation

In the final week, I wrapped up all my work into **reports, presentations, and video demonstrations**. I revised all my code, created visual summaries, and recorded short videos to explain my projects and what I learned from each one.

Looking back, this internship helped me:

- Understand how real-world AI/ML projects are structured
- Work with different types of data: tabular, text, images, and time-series
- Apply both basic and advanced machine learning techniques
- Improve my confidence in working independently with AI tools

From EDA and regression to BERT and CNNs, this journey was **intense, exciting, and incredibly rewarding**. I now feel ready to take on even more complex challenges in the AI/ML field.

Github link where all my work is uploaded:
https://github.com/TheBilalButt/Intern-at-DevelopersHub/tree/main

End of V10.
